Problem Set 1

Applied Stats/Quant Methods 1

Due: September 30, 2024

Instructions

- Please show your work! You may lose points by simply writing in the answer. If the problem requires you to execute commands in `R`, please include the code you used to get your answers. Please also include the `.R` file that contains your code. If you are not sure if work needs to be shown for a particular problem, please ask.
- Your homework should be submitted electronically on GitHub.
- This problem set is due before 23:59 on Monday September 30, 2024. No late assignments will be accepted.

Question 1: Education

A school counselor was curious about the average of IQ of the students in her school and took a random sample of 25 students' IQ scores. The following is the data set:

1. Find a 90% confidence interval for the average student IQ in the school.

   Based on the results of this analysis, we can say that we are 90 percent confident that the average IQ of the student population lies between 94.13 and 102.75. In order to arrive at this answer, I did the following:

   1. I calculated the sample mean which is 98.44
   2. I then calculate the sample standard deviation
   3. The standard deviation is approximately 13.09. This tells is the varience of our sample.
   4. I found the z-score associated with the 90 percent confidence interval which is 1.645. This value comes from the standard normal distribution, where 90 percent of

the distribution lies within 1.645 standard deviations from the mean whereby leaving 5 percent in each tail.

Please see the following code I used to arrive at this confidence interval.

```
#Problem Set 1 Jennie Moynihan

# Question 1: Find a 90% confidence interval for the average student IQ in the school.

# Load the data
Student_IQ <- c(105 , 69 , 86 , 100 , 82 , 111 , 104 , 110 , 87 , 108 , 87 , 90 , 94 , 113 , 112 , 98 ,
        80 , 97 , 95 , 111 , 114 , 89 , 95 , 126 , 98)
sum(Student_IQ)

# I get the sample mean
mean(Student_IQ)

# I calculate the standard deviation to measure the variability in the sample
sd(Student_IQ)
#The SD is approximately 13.09

# Then I calculate the standard error of the sample mean

# I next to calculte the z-score for the 90% confidence interval

qnorm(1 - (1 - 0.90) / 2)

# Now I need to use these values to calculate the lower bound and the upper bound of the confidence interval.

# Lower bound

lower_90 = (mean(Student_IQ)-(1.645)*(sd(Student_IQ)/sqrt(length((Student_IQ)))

# Answer = 94.13

# Upper Bound

upper_90 = (mean(Student_IQ)+(1.645)*(sd(Student_IQ)/sqrt(length((Student_IQ)))

# Answer = 102.74

lower_90
mean(Student_IQ)
upper_90

#Based on the results of this analysis, we can say that we are 90%
```

```
confident that the true population mean IQ lies between 94.132 and 102.7476

# Question 2: Next, the school counselor was curious whether the average student IQ in her school
```

2. Next, the school counselor was curious whether the average student IQ in her school is higher than the average IQ score (100) among all the schools in the country.

   Using the same sample, conduct the appropriate hypothesis test with $\alpha = 0.05$.

Based on the results, I think we can conclude that the p-value is 0.7215 which is greater than the significance level of 0.05 which means what we cannot reject the null hypothesis. Alternatively put, there is insufficient evidence to suggest that the school counselor is correct in her hypothesis that the mean IQ of students in their school is higher than that of the country. Instead, there is evidence to suggest that the mean IQ of the sample is lower than or equal to 100.

```
# Question 2: Next, the school counselor was curious whether the average student IQ in her school
#is higher than the average IQ score (100) among all the schools in the country.

# Using the same sample, conduct the appropriate hypothesis test with  = 0:05.

# To conduct the hypothesis test, I am going to follow five steps:

# Step 1: Check assumptions re data

# Student IQ is a continuous and numerical variable

# The sample size (n) is less than 30, we cannot involke CLT.

# We assume a randomized sample

#Step 2

# State hypotheses

# Set up the null (H0): The null hypothesis states that the average student IQ in the school is equal to or less than the county average (100).

# Alternative hypothesis (H1): The average student IQ is greater than the county average.

# Step 3: Calculate the test statistic

# Given that there is a strong directional expectation here, we require a one-tail test.

t.test (Student_IQ, mu=100, alternative = "greater")

# The Results of the One Sample t-test

#data:  Student_IQ
```

```
#t = -0.59574, df = 24, p-value = 0.7215
# hypothesis: true mean is greater than 100
#95 percent confidence interval:
#93.95993      Inf
#sample estimates:
#mean of x
#98.44
#data:  Student_IQ

#Step 5: Interpret the results

# Based on these results, I think we can conclude that the p-value is 0.7215, which is greater than the common significance level of 0.05. This means we fail to reject the null hypothesis.
# There is no evidence to suggest that the true mean IQ of the population is less than 100, based on this sample.We therefore cannot reject the null meaning that the sample mean is unlikelty to be higher than that of the true population.
```

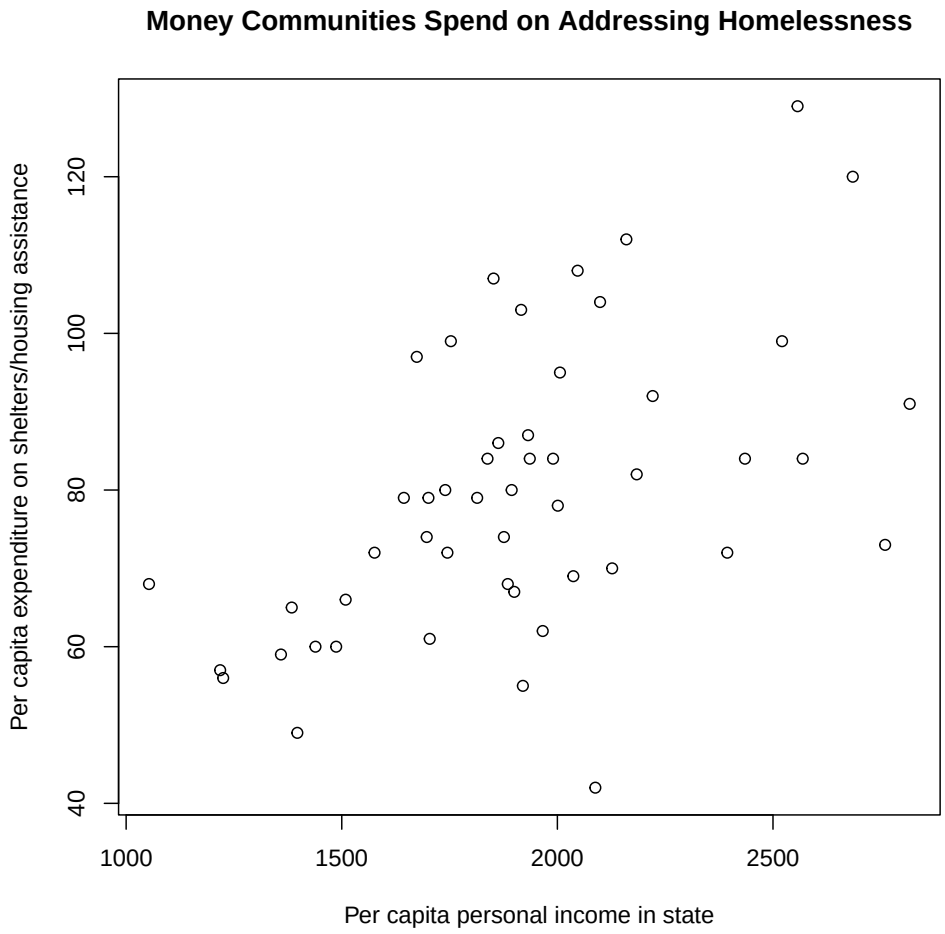
Question 2: Political Economy

Researchers are curious about what affects the amount of money communities spend on addressing homelessness. The following variables constitute our data set about social welfare expenditures in the USA.

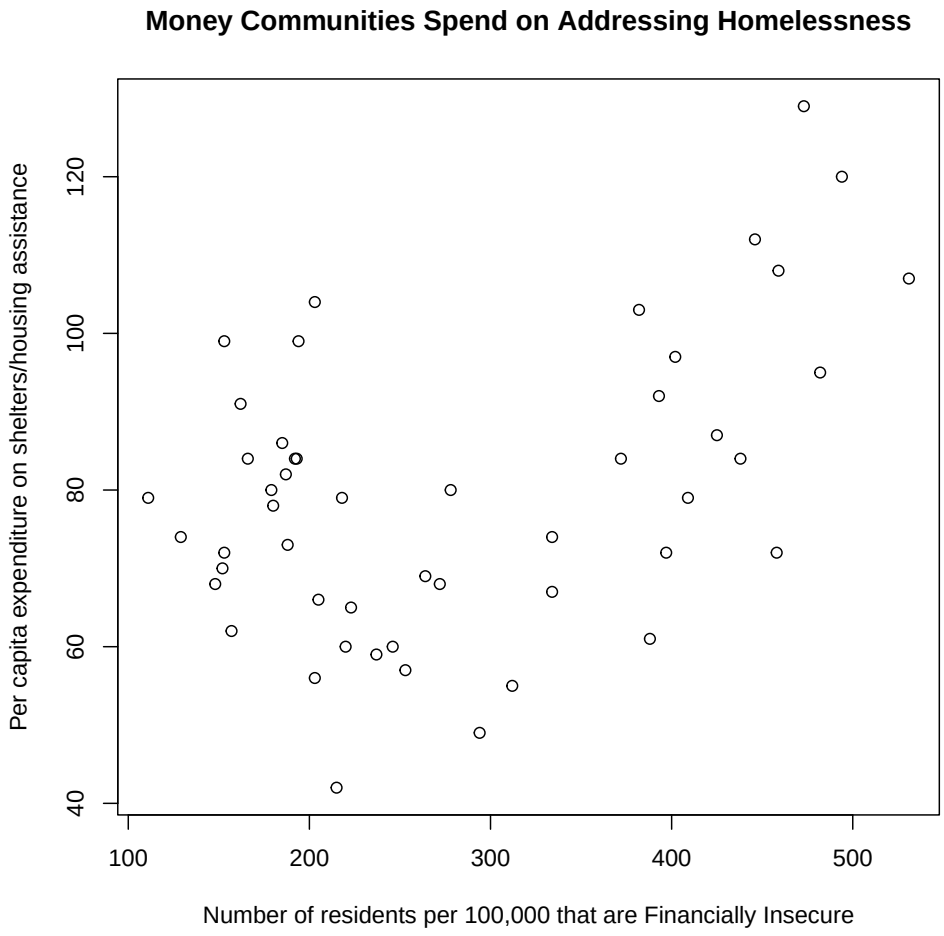
`State`	*50 states in US*
`Y`	*per capita expenditure on shelters/housing assistance in state*
`X1`	*per capita personal income in state*
`X2`	*Number of residents per 100,000 that are "financially insecure" in state*
`X3`	*Number of people per thousand residing in urban areas in state*
`Region`	*1=Northeast, 2= North Central, 3= South, 4=West*

Explore the `expenditure` data set and import data into `R`.

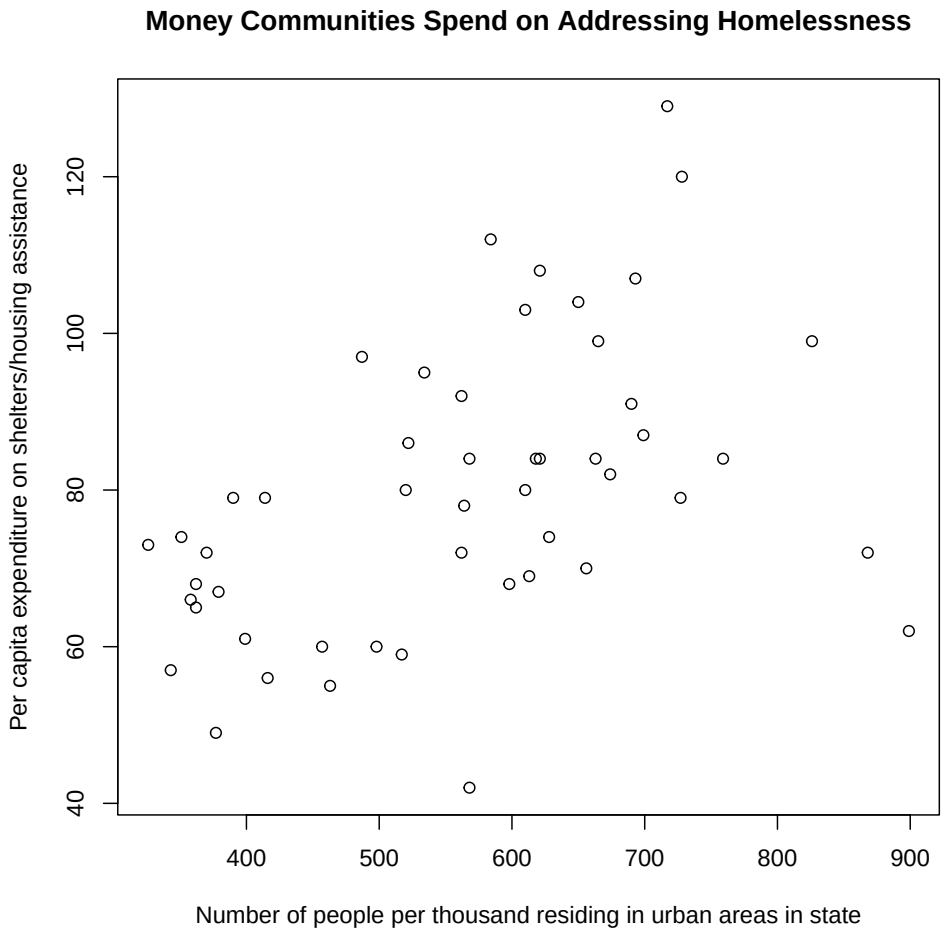
- Please plot the relationships among *Y*, *X1*, *X2*, and *X3*? What are the correlations among them (you just need to describe the graph and the relationships among them)?



This scatterplot shows us that there is a generally positive association between the per capita personal income and its associated spending on housing assistance (i.e., as personal income per capita increases, so too does the per capita expenditure on housing assistance). However, the observations are spread out, suggesting variability in how different states allocate resources for homelessness despite having similar income levels. This suggests that factors other than income alone influence expenditure levels.

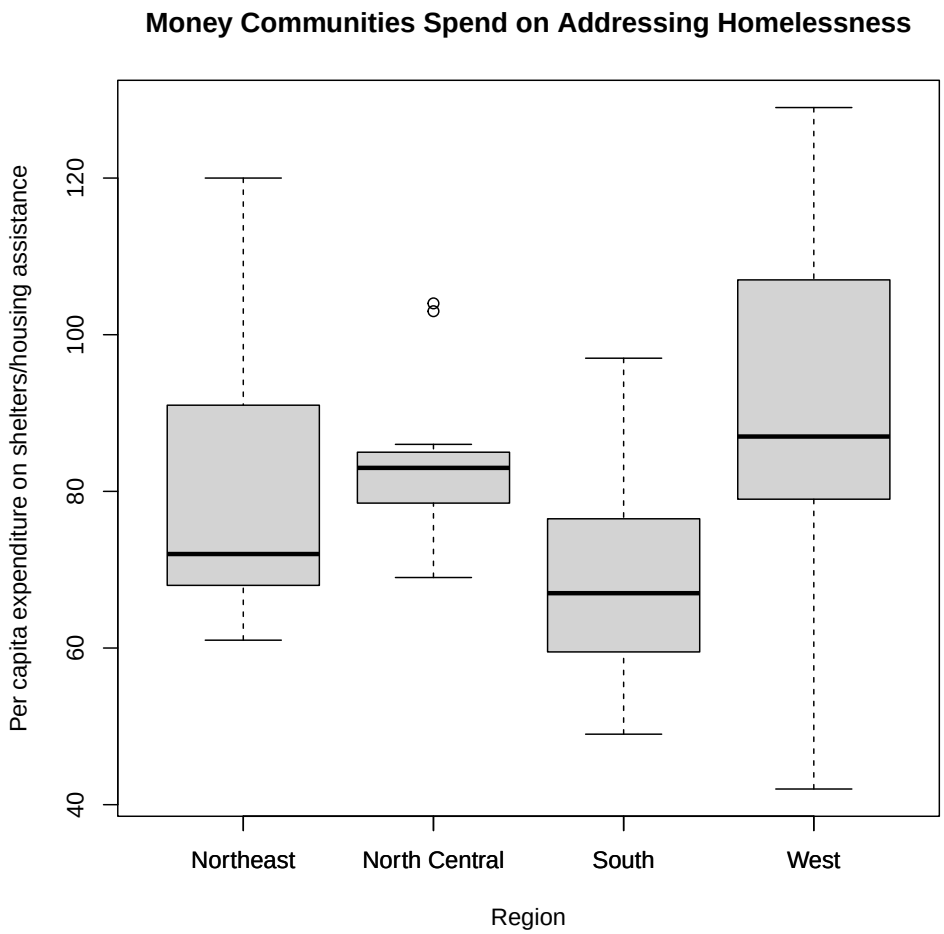


This scatterplot shows a somewhat unusual distribution and one that doesn't have a strong, directional association. Alternatively put, there doesn't appear to be an assocation between the number of a states inhabilitants that are financially insecure and its associated expenditure on housing assistance. Instead, there is a small cluster of states that have comparatively low numbers of financially insecure residents yet a comparatively high per capita expenditure on housing assistance.



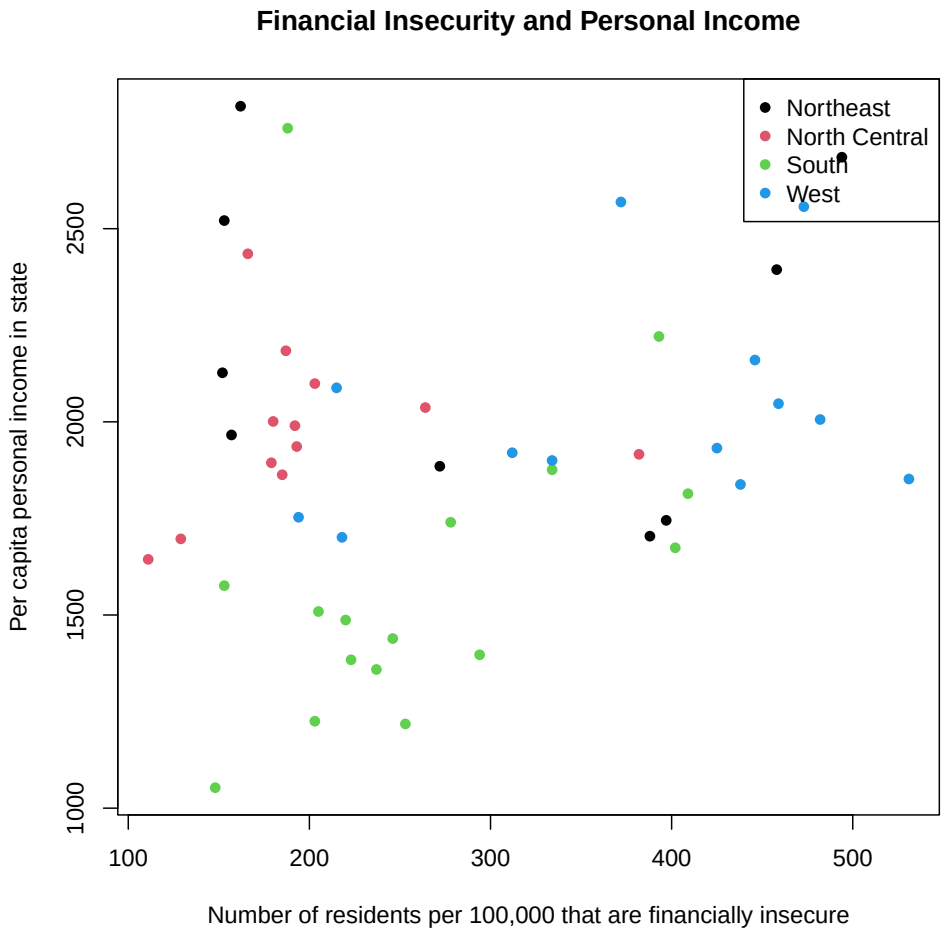
This scatterplot shows the number of people residing in urban areas in the state and the per capita expenditure on shelter/housing assistance. There appears to be a somewhat positive association between these two variables (i.e. states that have a greater number of urban residents generally have a slightly higher per capita expenditure on shelters and housing assistance) however there is a large amount of variability. There are also a small number of notable outliers, for example, there is one state with an urban resident population of approx 900 thousand with a per capita expenditure of only 64.

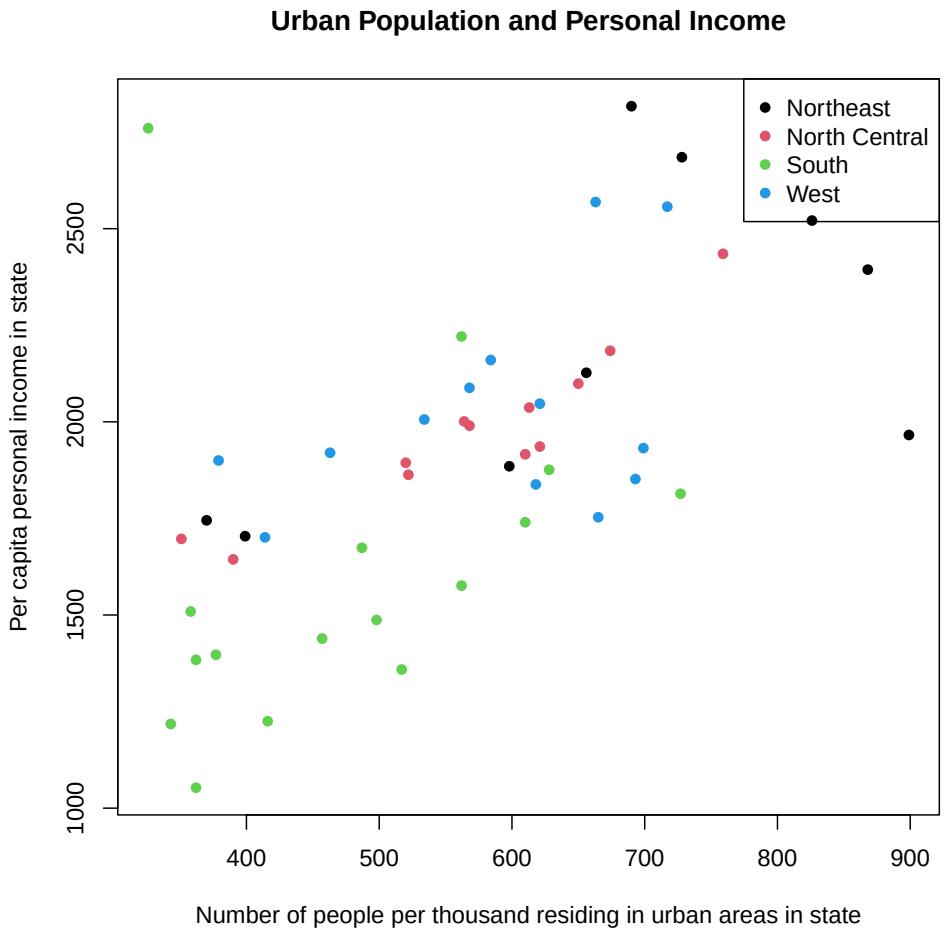
- Please plot the relationship between Y and *Region*? On average, which region has the highest per capita expenditure on housing assistance?

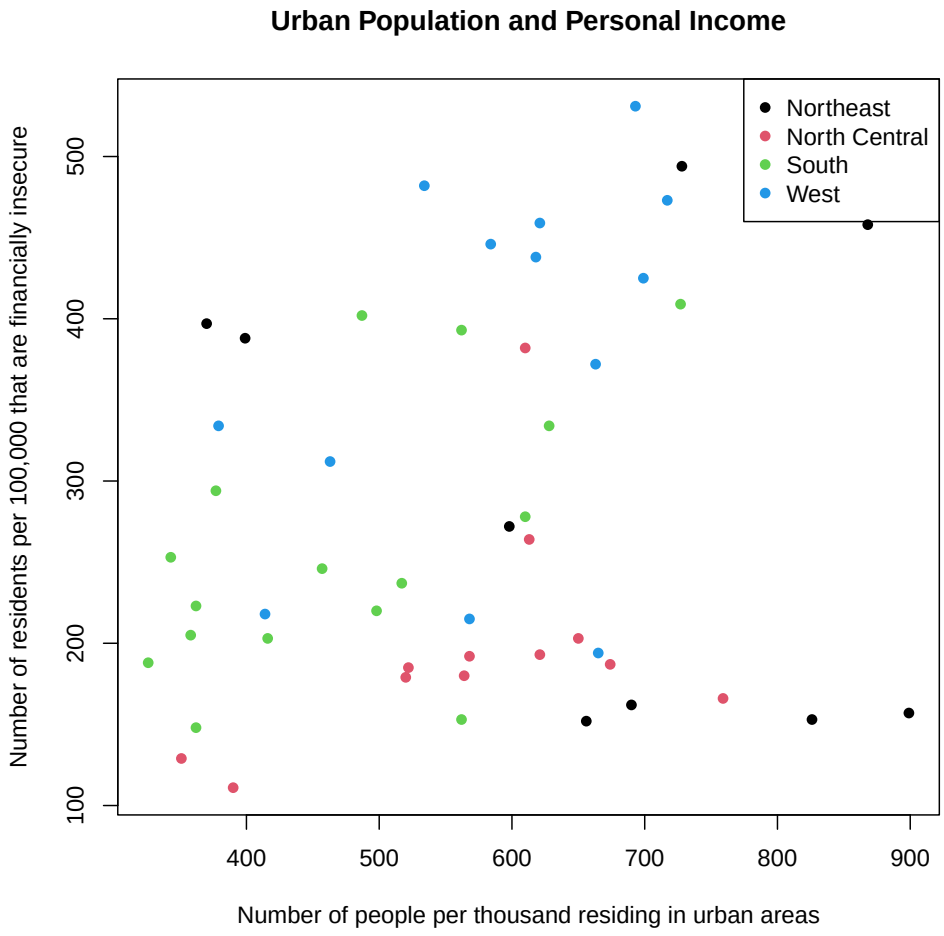


From the boxplot and assessing the median lines, it appears that the "West" has the highest per captia expenditure on housing assistance. However, the West also has the greatest spread. The "North Central" region has a similar average per capita expenditure but with a significally smaller spread.

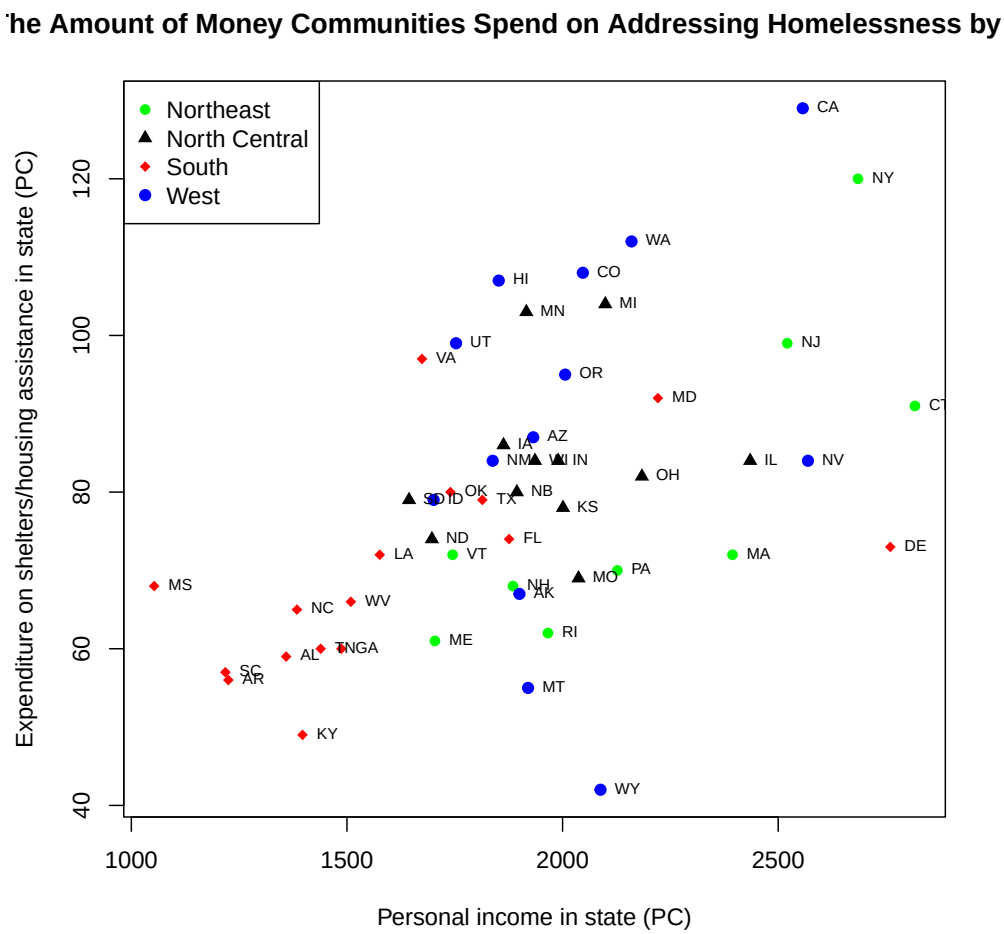
The following three graphs map the relationships between the three predictor variables.


Financial Insecurity and Personal Income
Northeast
North Central
South
West
Per capita personal income in state
1000
1500
2000
2500
100
200
300
400
500
Number of residents per 100,000 that are financially insecure


Urban Population and Personal Income
Northeast
North Central
South
West
Per capita personal income in state
1000
1500
2000
2500
400
500
600
700
800
900
Number of people per thousand residing in urban areas in state


Urban Population and Personal Income
Northeast
North Central
South
West
Number of residents per 100,000 that are financially insecure
100
200
300
400
500
400
500
600
700
800
900
Number of people per thousand residing in urban areas

- Please plot the relationship between Y and $X1$? Describe this graph and the relationship. Reproduce the above graph including one more variable *Region* and display different regions with different types of symbols and colors.



From this graph, we can broadly draw the following observations: It appears that the Southern regions of the USA generally have the lowest personal income per capita of the US states (there are exceptions, for example "DE" (Delaware) which is, for some reason, grouped as a Southern state. The states generally appear to have the highest expenditure per capita on housing assistance are those in the "West", "WY" (Wyoming) being a noteworthy outlier. Broadly speaking, we can observe the following from this data visualisation: there is a generally a positive association between per capita personal income and the associated expenditure on housing assistance programs.

```
# PART 2

# Load the data

```

```r
Expenditure <- read.table("https://raw.githubusercontent.com/ASDS-TCD/StatsI_Fall2024/main/datasets/expenditure.txt", header=T)

# I familiarise myself with the data

View(Expenditure)
head(Expenditure)
summary(Expenditure)

# I make the plots.

pdf("Y_X1PDF.pdf")
plot(y=Expenditure$Y,x=Expenditure$X1,
     ylab="Per capita expenditure on shelters/housing assistance",
     xlab="Per capita personal income in state",
      main="Money Communities Spend on Addressing Homelessness")
dev.off()

pdf("Y_X2PDF.pdf")
plot(y=Expenditure$Y,x=Expenditure$X2,
     ylab="Per capita expenditure on shelters/housing assistance",
     xlab="Number of residents per 100,000 that are Financially Insecure",
     main="Money Communities Spend on Addressing Homelessness")
dev.off()

pdf("Y_X3PDF.pdf")
plot(y=Expenditure$Y,x=Expenditure$X3,
     ylab="Per capita expenditure on shelters/housing assistance",
     xlab="Number of people per thousand residing in urban areas in state",
     main="Money Communities Spend on Addressing Homelessness")
dev.off()

# The wasn't happy with my few attempts at the following plot as it didn't clearly tell us anything (and it was hard to see the average). So I amended the command to make it a box plot.

pdf("Y_RegionPDF.pdf")
plot(y=Expenditure$Y,x=Expenditure$Region,
     ylab="Per capita expenditure on shelters/housing assistance",
     xlab="Region",
     main="Money Communities Spend on Addressing Homelessness")
axis(1, at = c(1, 2, 3, 4), labels = c("Northeast", "North Central", "South", "West"))
dev.off()

pdf("Y_X1RegionPDF.pdf")
plot(y=Expenditure$Y,x=Expenditure$X1,
     col=Expenditure$Region,
     pch=16,
```

```
     ylab="Expenditure on shelters/housing assistance in state (PC)",
     xlab="Personal income in state (PC)",
     main="Money Communities Spend on Addressing Homelessness by State")
dev.off()


pdf("X1_X2PDF.pdf")
plot(y=Expenditure$X1,x=Expenditure$X2,
     col=Expenditure$Region,
     pch=16,
     ylab="Per capita personal income in state",
     xlab="Number of residents per 100,000 that are financially insecure",
     main="Financial Insecurity and Personal Income")
legend("topright",
       legend = c("Northeast", "North Central", "South", "West"),
       col = 1:4,
       pch = 16)
dev.off()

# For these plots it made sense to include a key.

pdf("X1_X3PDF.pdf")
plot(y=Expenditure$X1,x=Expenditure$X3,
     col=Expenditure$Region,
     pch=16,
     ylab="Per capita personal income in state",
     xlab="Number of people per thousand residing in urban areas in state",
     main="Urban Population and Personal Income")
legend("topright",
       legend = c("Northeast", "North Central", "South", "West"),
       col = 1:4,
       pch = 16)
dev.off()

pdf("X2_X3PDF.pdf")
plot(y=Expenditure$X2,x=Expenditure$X3,
     col=Expenditure$Region,
     pch=16,
     ylab="Number of residents per 100,000 that are financially insecure",
     xlab="Number of people per thousand residing in urban areas",
     main="Urban Population and Personal Income")
legend("topright",
       legend = c("Northeast", "North Central", "South", "West"),
       col = 1:4,
       pch = 16)
dev.off()

# For the final graph, I defined the shapes and colours for each region.
```

```r

shapes <- c(16, 17, 18, 19)

colors <- c("green", "black", "red", "blue")

# I then created the final scatterplot using these different shapes and colors with a key in the left.

pdf("Final_PlotPDF.pdf")
plot(y = Expenditure$Y, x = Expenditure$X1,
     col = colors[Expenditure$Region],
     pch = shapes[Expenditure$Region],
     ylab = "Expenditure on shelters/housing assistance in state (PC)",
     xlab = "Personal income in state (PC)",
     main = "The Amount of Money Communities Spend on Addressing Homelessness by State")

legend("topleft",
       legend = c("Northeast", "North Central", "South", "West"),
       col = colors,
       pch = shapes)

# I also added the state names as labels next to each observation for clarify
text(x = Expenditure$X1, y = Expenditure$Y,
     labels = Expenditure$STATE,
     pos = 4,
     cex = 0.7)
dev.off()
```